

THE HEXAGONAL PRISM UNDER A CLOSED-NEIGHBOURHOOD ONE-ROUND RULE: FRACTIONAL ETERNAL DOMINATION NUMBER 46/13

ABSTRACT. For the hexagonal prism $C_6 \square K_2$, and for the *closed-neighbourhood* reading of the defender's one-round rule set out in Section 1,

$$\gamma_f^\infty(C_6 \square K_2) = \frac{46}{13} = 3.\overline{538461},$$

and the infimum is attained. That reading is ours and not a quotation: the source states the rule in a form we read as leaving unmoved weight in place, and both certificates below encode that reading, so if the intended rule is stricter they concern a different game and the value above is not the value asked for. This graph is the cell $k = 1$ of the first family of Problem 7.5 of Devvrit, Krim-Yee, Kumar, MacGillivray, Seamone, Virgile and Xu, who asked for the exact value of the fractional eternal domination number $\gamma_f^\infty(G)$ when G is a cubic abelian Cayley graph of the two exceptional types $C_{4k+2} \square K_2$ ($k \neq 2$) and $\text{Cay}(\mathbb{Z}_{8k}, \{\pm 1, 4k\})$, and who prove for this cell only $\frac{7}{2} < \gamma_f^\infty(C_6 \square K_2) \leq 4$. The upper bound here is an explicit 12-member family of rational weight functions, printed in full; the lower bound is a 9-node attack tree together with a dual certificate in exact rationals, also printed in full, so both halves can be checked against the printed objects alone — the 493 dual column inequalities being tedious rather than difficult. A closing remark records that, for the same reading of the rule, the last step of the published proof of $\frac{7}{2} < \gamma_f^\infty$ is not justified; the published inequality is nevertheless true, since $\frac{7}{2} < \frac{46}{13}$. No other cell of the authors' question is settled here, and we do not settle which rule they intended.

1. INTRODUCTION

We use the definitions of Devvrit et al. [1]. A *fractional dominating function* of a graph G is a map $w: V(G) \rightarrow \mathbb{R}_{\geq 0}$ with $\sum_{x \in N[v]} w(x) \geq 1$ for every $v \in V(G)$, and G is *S-fractionally dominated* if it admits a fractional dominating function of total weight at most S . In the eternal game, weights are assigned so that S -fractional domination is maintained under vertex attacks: writing w_i for the weight function at time step i and $m_{xy,i}$ for the weight moved from x to $y \in N(x)$ in round i , the defender may move simultaneously from as many vertices as desired subject to $\sum_{y \in N(x)} m_{xy,i} \leq w_i(x)$, and the resulting w_{i+1} must again S -fractionally dominate G and satisfy $w_{i+1}(v_i) \geq 1$, where v_i is the vertex attacked in round i . The *fractional*

2020 *Mathematics Subject Classification.* 05C69, 05C57, 05C25, 90C05.

Key words and phrases. fractional eternal domination, cubic abelian Cayley graph, prism, linear programming duality, exact value.

eternal domination number $\gamma_f^\infty(G)$ is the infimum of those S for which G can be eternally S -fractionally dominated.

Two features of this definition are used below, and the first of them is our reading of the source rather than a quotation from it. The budget constraint we have quoted is stated in [1] as $\sum_{y \in N(x)} m_{xy,i} \leq w_i(x)$, an open-neighbourhood sum bounded by the weight at x ; we read it as leaving unmoved weight where it is, so that a legal round transports weight along *closed* neighbourhoods with budget $w_i(x)$ at x . Everything below — the family of Section 2 and the linear program of Section 3 alike — encodes that reading, and if the intended rule is stricter (for instance if weight must leave x along open neighbourhoods) then both encode a different game and the value below is not the value asked for. We have not found in [1] an update equation for w_{i+1} that decides between the two readings, and we do not decide it here; that is why Theorem 1 below is stated for the closed-neighbourhood rule rather than as a determination of the cell the authors asked about. The accompanying program encodes the same reading, so no check in it can discriminate between the two. Second, in the fractional model of [1] the attacker is unrestricted, in particular it may attack a vertex that already carries weight.

Devvrit et al. prove that a cubic abelian Cayley graph G satisfies $\gamma_f^\infty(G) < \gamma(G)$ if and only if $G \cong C_{4k+2} \square K_2$ or $G \cong \text{Cay}(\mathbb{Z}_{8k}, \{\pm 1, 4k\})$ for some positive integer k , and compute γ_f^∞ exactly for all other cubic abelian Cayley graphs. For the two exceptional families they ask:

Problem 7.5 of [1]. Let k be a positive integer. Determine the exact value of $\gamma_f^\infty(G)$ when G is isomorphic to

- $C_{4k+2} \square K_2$ for $k \neq 2$;
- $\text{Cay}(\mathbb{Z}_{8k}, \{\pm 1, 4k\})$.

Ordering the cells of this question by $|V|$, the value is already a theorem of [1] at $|V| = 8$, namely $\gamma_f^\infty(\text{Cay}(\mathbb{Z}_8, \{\pm 1, 4\})) = \frac{8}{3}$, and at $|V| = 20$, namely $\gamma_f^\infty(C_{10} \square K_2) = \frac{28}{5}$ (the cell $k = 2$ excluded by the first bullet). The smallest cell it leaves open is therefore $k = 1$ of the first bullet, the hexagonal prism $C_6 \square K_2$ on 12 vertices, where [1] proves

$$\frac{7}{2} < \gamma_f^\infty(C_6 \square K_2) \leq 4.$$

The upper bound there is derived from the general estimate $\gamma_f^\infty(G) \leq (n + \kappa)/(\kappa + 1)$ of [1] for a graph of connectivity κ , which at $n = 12$, $\kappa = 3$ gives $\frac{15}{4}$; so [1] in fact also yields $\gamma_f^\infty(C_6 \square K_2) \leq \frac{15}{4} = 3.75$ here.

Theorem 1. *For the closed-neighbourhood one-round rule read off above, $\gamma_f^\infty(C_6 \square K_2) = \frac{46}{13}$, and the infimum is attained: an eternal strategy of total weight exactly $\frac{46}{13}$ exists.*

The value is consistent with the published statements about this cell: $\frac{7}{2} < \frac{46}{13} < \frac{15}{4} < 4$, with $\frac{46}{13} - \frac{7}{2} = \frac{1}{26}$; and $\frac{46}{13} - \gamma_f(C_6 \square K_2) = \frac{46}{13} - 3 = \frac{7}{13} < 1$,

as the corollary $\gamma_f^\infty - \gamma_f < 1$ of [1] for cubic abelian Cayley graphs demands. (Here $\gamma_f(C_6 \square K_2) = 3$: the uniform weight $\frac{1}{4}$ is fractionally dominating with total 3, and summing the twelve closed neighbourhood inequalities gives $4S \geq 12$.) Section 2 proves the upper bound and Section 3 the lower bound, closing with a remark on the published proof of $\frac{7}{2} < \gamma_f^\infty$; Section 4 states what is *not* settled.

Notation. Throughout, $G = C_6 \square K_2$ with vertex set $V = \mathbb{Z}_6 \times \{0, 1\}$, edges $(i, j) \sim (i \pm 1, j)$ and $(i, 0) \sim (i, 1)$. We also write $v = i + 6j$, so that $V = \{0, \dots, 11\}$ with

$$\begin{array}{llll} N[0] = \{0, 1, 5, 6\} & N[1] = \{0, 1, 2, 7\} & N[2] = \{1, 2, 3, 8\} & N[3] = \{2, 3, 4, 9\} \\ N[4] = \{3, 4, 5, 10\} & N[5] = \{0, 4, 5, 11\} & N[6] = \{0, 6, 7, 11\} & N[7] = \{1, 6, 7, 8\} \\ N[8] = \{2, 7, 8, 9\} & N[9] = \{3, 8, 9, 10\} & N[10] = \{4, 9, 10, 11\} & N[11] = \{5, 6, 10, 11\}. \end{array}$$

Thus G is 3-regular with 18 edges, 3-connected, bipartite and of girth 4.

2. THE UPPER BOUND

Lemma 2. *Let $\mathcal{F} = \{f_v : v \in V\}$ be a family of fractional dominating functions of G , indexed by V , such that*

- (i) *each f_v has total weight S ;*
- (ii) *$f_v(v) = 1$ for every v ;*
- (iii) *for every ordered pair $(u, v) \in V \times V$ there is a transportation plan $x : \{(a, b) : b \in N[a]\} \rightarrow \mathbb{R}_{\geq 0}$ with $\sum_{b \in N[a]} x(a, b) = f_u(a)$ for all a and $\sum_{a: b \in N[a]} x(a, b) = f_v(b)$ for all b .*

Then $\gamma_f^\infty(G) \leq S$, and the infimum is attained.

Proof. Play from f_{v_0} for any v_0 , and respond to an attack at v , from the current configuration f_u , by moving to f_v ; (iii) says the move is legal for the one-round rule, since weight sent from a to a itself is weight left in place, (i) says the total is unchanged, f_v is fractionally dominating, and (ii) says $f_v(v) = 1 \geq 1$. The strategy therefore survives every infinite sequence of attacks at total weight exactly S . $\square$

For $(m, t) \in V$ put $f_{(m, t)}(i, j) = h_{j \oplus t}((i - m) \bmod 6)$, where $\oplus$ is addition mod 2 and

$$\begin{array}{llll} h_0 : 0 \mapsto 1, & \pm 1 \mapsto \frac{1}{13}, & \pm 2 \mapsto \frac{1}{13}, & 3 \mapsto \frac{5}{13}; \\ h_1 : 0 \mapsto \frac{6}{13}, & \pm 1 \mapsto 0, & \pm 2 \mapsto \frac{6}{13}, & 3 \mapsto \frac{6}{13}. \end{array}$$

Both profiles are even, so each $f_{(m, t)}$ is invariant under the stabiliser $\{\text{id}, i \mapsto -i\}$ of $(0, 0)$ and the twelve translates are well defined. Written out, the seed $f = f_{(0, 0)}$ is

$$f = \left(1, \frac{1}{13}, \frac{1}{13}, \frac{5}{13}, \frac{1}{13}, \frac{1}{13} \mid \frac{6}{13}, 0, \frac{6}{13}, \frac{6}{13}, \frac{6}{13}, 0\right)$$

in the order $v = 0, \dots, 11$, of total weight $1 + \frac{9}{13} + \frac{24}{13} = \frac{46}{13}$. Its twelve closed neighbourhood sums are

$$\frac{21}{13}, \frac{15}{13}, 1, 1, 1, \frac{15}{13} \mid \frac{19}{13}, 1, 1, \frac{23}{13}, 1, 1,$$

all at least 1; translation carries this to every member, so (i) and (ii) of Lemma 2 hold with $S = \frac{46}{13}$.

For (iii), here is one plan in full, for $f_{(0,0)} \rightarrow f_{(3,0)}$; the notation $a \rightarrow b = q$ means that a ships q to b , and $b \in N[a]$ in every case:

$$\begin{array}{lllll} 0 \rightarrow 0 = 5/13 & 0 \rightarrow 1 = 1/13 & 0 \rightarrow 5 = 1/13 & 0 \rightarrow 6 = 6/13 & 1 \rightarrow 2 = 1/13 \\ 2 \rightarrow 3 = 1/13 & 3 \rightarrow 3 = 5/13 & 4 \rightarrow 3 = 1/13 & 5 \rightarrow 4 = 1/13 & 6 \rightarrow 7 = 6/13 \\ 8 \rightarrow 9 = 6/13 & 9 \rightarrow 3 = 6/13 & 10 \rightarrow 11 = 6/13 & & \end{array}$$

Every vertex ships exactly its own weight and every vertex receives exactly the target weight,

$$\begin{array}{l} \text{ships : } (1, \frac{1}{13}, \frac{1}{13}, \frac{5}{13}, \frac{1}{13}, \frac{1}{13} \mid \frac{6}{13}, 0, \frac{6}{13}, \frac{6}{13}, \frac{6}{13}, 0), \\ \text{receives : } (\frac{5}{13}, \frac{1}{13}, \frac{1}{13}, 1, \frac{1}{13}, \frac{1}{13} \mid \frac{6}{13}, \frac{6}{13}, 0, \frac{6}{13}, 0, \frac{6}{13}), \end{array}$$

the second of which is $f_{(3,0)}$. The total shipped is $\frac{46}{13}$, and the attacked vertex 3 receives $\frac{1}{13} + \frac{5}{13} + \frac{1}{13} + \frac{6}{13} = 1$. All 144 ordered pairs of the family are feasible for (iii); each is a transportation problem with integer data after scaling by 13, and the accompanying program decides all 144 by exact integer maximum flow.

By Lemma 2, $\gamma_f^\infty(C_6 \square K_2) \leq \frac{46}{13}$.

3. THE LOWER BOUND

3.1. A finite attack tree is a relaxation. Fix a finite rooted tree T with nodes $0, 1, \dots, N$, root 0, parent map p , and an attacked vertex $a_k \in V$ for each $k \geq 1$. Consider

$$\begin{array}{ll} \text{minimise } S & \\ \text{subject to } \sum_{v \in V} w_k(v) = S & (0 \leq k \leq N) \\ \sum_{u \in N[v]} w_k(u) \geq 1 & (0 \leq k \leq N, v \in V) \\ (1) \quad w_k(a_k) \geq 1 & (1 \leq k \leq N) \\ \sum_{y \in N[u]} x_k(u, y) = w_{p(k)}(u) & (1 \leq k \leq N, u \in V) \\ \sum_{u: y \in N[u]} x_k(u, y) = w_k(y) & (1 \leq k \leq N, y \in V) \\ S \geq 0, w_k(v) \geq 0, x_k(u, y) \geq 0. & \end{array}$$

Lemma 3. *If L is the optimum of (1) then $L \leq \gamma_f^\infty(G)$.*

Proof. An eternal defender surviving at total weight S survives, in particular, the attacks of T : taking w_0 to be its initial configuration, w_k the configuration reached along the path from the root to k , and x_k the moves it makes in that round, every constraint of (1) is one of the game's own requirements. So S is feasible for (1) and $L \leq S$; take the infimum over S . $\square$

Three remarks make Lemma 3 robust rather than delicate. The root configuration w_0 is *free*, so no uniqueness or symmetry lemma about optimal initial configurations is needed. Any subtree of a legal attacker is a legal attacker, so pruning T can only decrease L — it can never invalidate the bound. And an attacker that plays one fixed first attack and only then branches is weaker than one that branches immediately, so using such a tree is safe.

3.2. The tree. Take $N = 8$ with

$$\begin{aligned} p(1) &= 0, & a_1 &= (0, 0) \\ p(2) &= 1, \ p(4) = 1, \ p(7) = 1, & a_2 &= (2, 1), \ a_4 = (3, 1), \ a_7 = (4, 1) \\ p(3) &= 2, \ p(5) = p(6) = 4, \ p(8) = 7, & a_3 &= a_6 = a_8 = (5, 1), \ a_5 = (1, 1). \end{aligned}$$

The second attacks $\{(2, 1), (3, 1), (4, 1)\}$ are the three vertices of the other hexagon at distances 3, 4, 3 from the first attack. For this tree (1) has 493 variables (S ; the 9×12 weights $w_k(v)$; and 8×48 flows $x_k(u, y)$, one per node $k \geq 1$ and per ordered pair (u, y) with $y \in N[u]$), 201 equalities (nine total-weight rows, then $8 \times (12 + 12)$ flow rows) and 116 inequalities (9×12 domination rows and 8 attack rows).

Proposition 4. *The optimum of (1) for this tree is exactly $\frac{46}{13}$; consequently $\gamma_f^\infty(C_6 \square K_2) \geq \frac{46}{13}$.*

Proof. Only “ $\geq$ ” is needed for the theorem, and it follows from weak duality once a dual feasible point of objective $\frac{46}{13}$ is exhibited. Let $\alpha_k \geq 0$ be the multiplier of the row $w_k(a_k) \geq 1$, let $\beta_{k,v} \geq 0$ be the multiplier of the row $\sum_{u \in N[v]} w_k(u) \geq 1$, and let $\lambda_k, \mu_{k,u}^{\text{out}}, \mu_{k,y}^{\text{in}}$ be the free multipliers of the total-weight, out-flow and in-flow equalities respectively. Dual feasibility is the system of 493 column inequalities of (1), one per primal variable, and the dual objective is $\sum_k \alpha_k + \sum_{k,v} \beta_{k,v}$.

Take $\alpha_1 = \frac{4}{13}, \alpha_2 = \frac{2}{13}, \alpha_4 = \frac{4}{13}, \alpha_5 = \frac{3}{13}, \alpha_6 = \frac{2}{13}, \alpha_7 = \frac{2}{13}, \alpha_3 = \alpha_8 = 0$, and let $13\beta_{k,v}$ be given by

$$\begin{aligned} k = 1 : & (1, 2) \mapsto 1, (1, 3) \mapsto 1, (1, 4) \mapsto 1, (1, 7) \mapsto 2, \\ & (1, 8) \mapsto 1, (1, 10) \mapsto 1, (1, 11) \mapsto 2 \\ k = 2 : & (2, 1) \mapsto 1, (2, 3) \mapsto 1, (2, 11) \mapsto 1 \\ k = 4 : & (4, 1) \mapsto 1, (4, 2) \mapsto 2, (4, 4) \mapsto 2, (4, 5) \mapsto 1, \\ & (4, 6) \mapsto 1, (4, 11) \mapsto 1 \\ k = 5 : & (5, 0) \mapsto 1, (5, 2) \mapsto 1, (5, 10) \mapsto 1 \\ k = 6 : & (6, 0) \mapsto 1, (6, 4) \mapsto 1, (6, 8) \mapsto 1 \\ k = 7 : & (7, 3) \mapsto 1, (7, 5) \mapsto 1, (7, 7) \mapsto 1, \end{aligned}$$

all other $\beta_{k,v}$ being 0. There are 31 nonzero multipliers on the $\geq$ rows; the attack part sums to $(4 + 2 + 4 + 3 + 2 + 2)/13 = \frac{17}{13}$ and the domination part to $(9 + 3 + 8 + 3 + 3 + 3)/13 = \frac{29}{13}$, so the dual objective is exactly $\frac{17}{13} + \frac{29}{13} = \frac{46}{13}$.

The free multipliers are $\lambda_3 = -\frac{7}{13}, \lambda_4 = -\frac{1}{13}, \lambda_5 = -\frac{3}{13}, \lambda_6 = -\frac{2}{13}$, all other $\lambda_k = 0$, together with the following, where each row lists $13\mu_{k,u}^{\text{out}}$ for

$u = 0, 1, \dots, 11$ and then $13\mu_{k,y}^{\text{in}}$ for $y = 0, 1, \dots, 11$; all multipliers not listed — in particular every flow multiplier of nodes 1 and 8 — are 0:

k=2 out:	6	5	5	5	6	6	6	5	5	5	6	6
k=2 in:	-6	-6	-5	-6	-6	-6	-6	-5	-6	-6	-6	-6
k=3 out:	7	7	7	7	7	7	7	7	7	7	7	7
k=3 in:	-7	-7	-7	-7	-7	-7	-7	-7	-7	-7	-7	-7
k=4 out:	-1	-1	-1	0	-1	-1	-1	-1	0	0	0	-1
k=4 in:	1	1	0	0	0	1	1	0	0	0	0	0
k=5 out:	1	0	1	2	2	2	0	0	0	2	2	2
k=5 in:	-2	-1	-2	-2	-2	-2	-2	0	-2	-2	-2	-2
k=6 out:	0	1	1	1	0	0	0	1	1	1	0	0
k=6 in:	-1	-1	-1	-1	-1	0	-1	-1	-1	-1	-1	0
k=7 out:	-1	-1	-1	-2	-2	-2	-1	-1	-1	-2	-2	-2
k=7 in:	1	1	1	1	2	1	1	1	1	1	2	1

This is 123 nonzero equality multipliers, every one of denominator 13. Substituting into the 493 column inequalities of (1) verifies them all in exact rational arithmetic; the accompanying program does exactly that, having first rebuilt (1) from the description above. Weak duality now gives $L \geq \frac{46}{13}$, and Section 2 together with Lemma 3 gives $L \leq \gamma_f^\infty(G) \leq \frac{46}{13}$, so $L = \frac{46}{13}$ as well. $\square$

Theorem 1 follows from Section 2 and Proposition 4.

Remark. The inequality $\frac{7}{2} < \gamma_f^\infty(C_6 \square K_2)$ is true, by Theorem 1, but its published proof ends with a step that is not justified for the reading of the rule used here; we record this because a reader reconstructing the lower bound from [1] will otherwise reconstruct it. In the labelling of the relevant figure of [1] — inner cycle $abcdef$, outer cycle $ghijkl$, with $a \sim g$, $b \sim h$, $\dots$, $f \sim l$ — the argument assumes a first attack at a and derives a family of configurations parameterised by $x \in [0, \frac{1}{2}]$, namely $a = 1$, $b = f = h = l = 0$, $c = e = x$, $d = j = \frac{1}{2} - x$, $g = i = k = \frac{1}{2}$. Its last step asserts that, after a response to a second attack at j , “there must be a weight of 1 on j , $\frac{1}{2}$ on d and 0 on each of the vertices c, e, h, i, k, l ”. For the closed-neighbourhood reading used throughout this note that is not forced: start from the $x = 0$ member

$$a = 1, \quad d = g = i = j = k = \frac{1}{2}, \quad b = c = e = f = h = l = 0$$

of total weight $\frac{7}{2}$, and answer an attack at j by the one-round move

$$\begin{aligned} a \rightarrow b = \frac{1}{2}, \quad a \rightarrow f = \frac{1}{2}, \quad g \rightarrow a = \frac{1}{4}, \\ i \rightarrow h = \frac{1}{4}, \quad i \rightarrow j = \frac{1}{4}, \quad k \rightarrow l = \frac{1}{4}, \quad k \rightarrow j = \frac{1}{4}, \end{aligned}$$

with d, j and the remaining $\frac{1}{4}$ at g left in place. No vertex ships more than it holds, the total is conserved, and the resulting configuration

$$\begin{aligned} a = \frac{1}{4}, \quad b = \frac{1}{2}, \quad c = 0, \quad d = \frac{1}{2}, \quad e = 0, \quad f = \frac{1}{2}, \\ g = \frac{1}{4}, \quad h = \frac{1}{4}, \quad i = 0, \quad j = 1, \quad k = 0, \quad l = \frac{1}{4} \end{aligned}$$

puts weight 1 on the attacked vertex j and has all twelve closed neighbourhood sums at least 1. So the attack at j is answerable at total weight $\frac{7}{2}$ from that configuration, and $h = l = \frac{1}{4} \neq 0$. The accompanying program checks this configuration and this response arc by arc. Whether the step can be justified under a stricter reading of the movement rule we do not say.

4. WHAT IS NOT SETTLED

- (1) **Which game is being played is not settled.** The value $\frac{46}{13}$ is a value for the closed-neighbourhood one-round rule of Section 1, which is our reading of [1] and not a quotation from it. Both certificates above and the accompanying program encode that reading, so none of them discriminates between it and a stricter one; for a stricter rule nothing here bounds γ_f^∞ , and the cell of the authors' question stays open.
- (2) **The authors' question is not determined.** At most one cell of it is. If the reading above is the intended one, the cells left open after Theorem 1 are $\{C_{4k+2} \square K_2 : k \geq 3\} \cup \{\text{Cay}(\mathbb{Z}_{8k}, \{\pm 1, 4k\}) : k \geq 2\}$: every cell of both families except $\frac{46}{13}$ and $\frac{28}{5}$ at $k = 1, 2$ of the first and $\frac{8}{3}$ at $k = 1$ of the second. Nothing here should be read as settling the question as a whole.
- (3) **No general theorem follows.** The method is cell by cell: an explicit finite family of weight functions, and a finite attack tree. It gives no construction for general k , and it does not address the authors' separate questions of whether γ_f^∞ is always rational, or whether n fractional dominating functions always suffice — the latter is what makes a bounded search for a family inconclusive rather than a lower bound when it finds nothing.
- (4) **Only the 9-node dual is certified here.** Other attack trees were solved in the course of the work; none of their optima is quoted or used above, and no certificate for any of them is printed. Theorem 1 rests on the 9-node dual printed above and on nothing else.
- (5) **The certificates were found numerically.** A floating-point linear programming solver located the witness family and the dual vector. That is how they were found and not why they hold: every object above is exhibited as exact rationals and every verification — fractional domination, transportation feasibility, the 493 dual column inequalities, the objective — is carried out in exact integer or rational arithmetic. No floating-point value appears anywhere in the chain of reasoning.

5. REPRODUCTION AND VERIFICATION

Everything needed is printed above: the family is determined by the profiles h_0, h_1 of Section 2; the lower bound by the attack tree, the linear program (1) with its stated row and variable ordering, and the $31 + 123$

multipliers of Proposition 4. Reconfigurability of a single ordered pair is a transportation problem with integer data after scaling by 13, and the plan printed in Section 2 can be checked by hand; the dual objective $\frac{17}{13} + \frac{29}{13}$ can be re-added by hand; the 493 dual column inequalities are the one part that is tedious rather than difficult.

A program accompanying this note re-derives all of it from the printed objects, in exact integer and **Fraction** arithmetic, using only the Python standard library: it rebuilds G and checks it against the closed neighbourhoods printed above; rebuilds the family from its profiles and checks total weight, fractional domination and the centre condition for all 12 members; checks the printed transportation plan arc by arc; decides all 144 ordered pairs by exact integer maximum flow; rebuilds (1) and verifies its dimensions 493/201/116, then verifies all 493 dual column inequalities and the dual objective; and re-checks the configuration and the response of the remark that closes Section 3. What it does not do is re-run the floating-point linear programs that found the family and the dual vector; those are re-checked only as exhibited objects. Nor can it bear on the reading of the movement rule: it encodes the same reading, in the same closed neighbourhoods.

The output of the recorded run is wider than the note: besides the checks just listed it records controls, a second family of weight functions of total weight $\frac{46}{13}$, three further transportation plans, an ancillary upper bound for $\text{Cay}(\mathbb{Z}_{16}, \{\pm 1, 8\})$, and a closing scope note whose phrase “the accompanying note” should not be read as a scope statement about the sections above. Nothing above depends on any of that extra material, and none of it is claimed here; what is and is not settled here is stated in Section 4.

REFERENCES

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